

CSCE 640 QUANTUM ALGORITHMS · MODULE 13 · LECTURE NOTES

Claiming a Quantum Advantage

The problem, the baseline, and the machine.

MODULE OUTCOMES

- State what an advantage claim must specify.
- Identify the standard overclaims.
- Explain the dequantisation pattern.
- Place this course in the semester and the program.
- Assess a quantum computing claim in minutes.

1 What a claim must name

AN ADVANTAGE CLAIM NAMES A PROBLEM, A CLASSICAL BASELINE, AND A MACHINE — AND ALL THREE ARE LOAD-BEARING

The problem — and whether anybody wants it solved: random circuit sampling is a real and impressive demonstration and is not a useful problem, which the careful papers state plainly and the coverage routinely omits (Module 11 §4).

The classical baseline — which algorithm, running on what hardware, with how much engineering effort spent on it — because the baseline is the claim's other half, and it is frequently a strawman or simply the first thing that came to hand (§2).

The machine — qubit count, gate error rates, connectivity, and whether error correction was assumed rather than performed (Module 11 §1) — since a result assuming error-free qubits is a mathematical result rather than an engineering one, and both are legitimate but they are not the same.

And the resource count for a useful instance (Module 08 §4 and Module 10 §4) — which converts a possibility into a number that somebody else can check, and is the single most useful thing a paper in this area can include.

The claim	The correction
'Exponentially faster.'	At what task? Than which classical method? And is it queries or time? (Module 05 §4.)
'It tries all the answers at once.'	Which predicts constant-time unstructured search, and that is provably false (Module 01 §2, Module 06 §4).

The claim	The correction
'It will break all encryption.'	Public-key number-theoretic primitives, not symmetric ones; and not on any machine that exists (Module 08 §3).
'Quantum speedup for machine learning.'	Input and output costs, almost always unaccounted for (Module 11 §3).
'Solved in 200 seconds what would take 10,000 years.'	Versus which classical implementation, run for how long, by people with what incentive to try? (§2.)
'Quantum supremacy achieved.'	On a sampling task constructed specifically to be hard to simulate — which is the experiment's design, not a criticism of it.

The baseline is the half of the claim nobody checks — and it is the half that has moved most often after publication (§2). Every row's correction above is about something the claim did not name, which is why §4's checklist is a list of questions rather than a list of objections: the problem is almost never that a stated fact is wrong.

2 Dequantisation

SEVERAL CLAIMED ADVANTAGES HAVE BEEN REDUCED OR ELIMINATED BY BETTER CLASSICAL ALGORITHMS WRITTEN AFTERWARDS

A quantum algorithm is published with an apparent exponential speedup over the known classical method, and a classical algorithm matching it, or very nearly, follows — sometimes within months, and in several well-known cases by a student.

Because the quantum algorithm's assumptions frequently imply classical structure as well — a sampling-access assumption strong enough to make the quantum version work often enables a classical sampling algorithm too, which is the mechanism behind the quantum-machine-learning dequantisation results.

And several sampling demonstrations have had their claimed classical cost reduced by many orders of magnitude after the fact, by better tensor-network simulation and by simply spending more compute on the classical side than the original comparison did.

Which is science working — the claim provoked exactly the effort that tested it — and it is a strong argument for stating the baseline precisely enough to be attacked, rather than vaguely enough to be safe. A claim nobody can refute is a claim nobody can confirm.

3 The semester

Course	Who sets the rules	The consequence
CSCE 640 (this one)	Physics.	The gate set, the measurement rule, and the noise are given — you design within them and cannot negotiate.
CSCE 628	Biology.	The data is noisy and the ground truth is frequently not available — so the algorithm's output cannot be checked the way a sorting routine's can.
CSCE 717	Self-interested agents.	The inputs are chosen to benefit whoever sent them — so correctness has to include the incentive to report honestly.

In all three courses the model comes from outside computer science — and designing against a model you did not choose is the semester's common skill. It is a different discipline from the earlier semesters, where the machine, the data, and the inputs were all yours: here the rules are given, they are not negotiable, and the first job in each case is to state them accurately. **The model-from-outside framing is Semester 12's result**, and Module 01 §4's insistence on resource counts is one instance of it.

4 Assessing a claim

- What is the problem, and does anybody want it solved? — **which eliminates the sampling demonstrations from the 'useful' column immediately**, without diminishing them as experiments.
- What is the classical baseline, and how hard did they try? (§2) — **and is it the best known method or merely the obvious one?** This is where most claims fail.
- What machine, with what error rates and connectivity, and is fault tolerance assumed or achieved? (Module 11 §1.)
- What is the resource count for a useful instance? (Module 08 §4, Module 10 §4) — **in physical qubits, not logical ones**, since the two differ by three orders of magnitude.
- And is the scaling measured or asserted? (Module 11 §4) — **over what range of problem sizes, and does that range actually support the fit? Five questions, five minutes — and most claims fail on the second one, which is the baseline nobody checks.**

WHERE THIS COURSE LEAVES YOU

You know the mechanism — interference, and the question of what cancels (Module 04 §4) — which is enough to evaluate whether a proposed algorithm could possibly work, and that filter alone resolves a great deal.

You can derive the important algorithms rather than recall them — Grover as a rotation built from two reflections, Shor as phase estimation on a modular multiplication operator — which is the difference between understanding and remembering.

And you know the engineering reality and the complexity-theoretic status (Modules 10 and 12) — which is what separates assessing this field from admiring it, and is the part that most popular accounts leave out entirely.

The closing rule is the program's, unchanged across thirty-four courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it is the problem, the baseline, and the machine — because 'a quantum computer did it faster' is not yet a claim, and the three missing names are where the content is.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Aaronson — Read the fine print (free)	§§1 and 2 — the caveats, stated once and carefully, and the right thing to read before assessing any proposal.
Tang — A quantum-inspired classical algorithm for recommendation systems (free)	§2's dequantisation — the paper that started the pattern, and it is short.
Pan, Chen & Zhang — classical simulation of random circuit sampling (free preprints)	§2's second kind — a claimed classical cost reduced by orders of magnitude after the fact.
Quantum (the journal), the resource-estimate papers (free)	§§1 and 4 — the papers that do state all three things, which are the model to imitate in Project 2.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 State the three things a claim must name**, and why each matters.
- 2 Find three published claims** and check which of the three each names.

- 3 **Correct six overclaims** in your own words.
- 4 **Read one dequantisation paper** and state what assumption it exploited.
- 5 **Find a sampling demonstration** and the classical response to it.
- 6 **Write the baseline section** for your own Project 2.
- 7 **Convert one logical-qubit estimate** to physical qubits, with the overhead stated.
- 8 **State each Semester 12 course's model and who sets it.**
- 9 **Run §4's five questions** on a current news story.
- 10 **Project 2 is now due.** Submit the derivation, the implementation, the best-known classical baseline measured on the same instances, the measured scaling of both, the resource estimate in physical qubits, and the honest claim naming the problem, the baseline, and the machine.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the three things an advantage claim must specify.
- 2 What fourth thing converts a possibility into a number?
- 3 Correct six standard overclaims.
- 4 Which half of a claim is least checked?
- 5 Describe the dequantisation pattern and its mechanism.
- 6 Why is it science working rather than scandal?
- 7 Give the five assessment questions in order.
- 8 Which question do most claims fail?
- 9 State each Semester 12 course's external model.
- 10 State the closing rule in this subject's terms.